Assignment - 2 Conversational AI

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Hybrid Retrieval-Augmented Generation (RAG) System on Wikipedia

This project implements a **Hybrid RAG system** combining **Dense Retrieval**, **Sparse Retrieval (BM25)**, and **Reciprocal Rank Fusion (RRF)** over a dynamically constructed Wikipedia corpus.
It also includes **automated evaluation**, **ablation studies**, and **LLM-based judging**.

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1. Dataset Requirements & Corpus Construction

The corpus consists of **500 Wikipedia articles**:

- **200 Fixed URLs** (constant across runs)
- **300 Random URLs** (sampled per indexing run)

Each article is cleaned, chunked, embedded, and indexed for retrieval.

Corpus Pipeline

Script	Description
`src/corpus/url_sampling.py`	Connects to Wikipedia and samples **200 fixed URLs** (currently configurable; default limit was 20)
`src/corpus/fetch_wikipedia.py`	Fetches Wikipedia page content and stores raw text with page titles
`src/corpus/clean_text.py`	Performs basic text cleaning
`src/corpus/chunker.py`	Sentence-based chunking (200–400 tokens with overlap) and metadata generation for BM25
`src/corpus/clean_text.py --input data_random/cleaned_text --output data_random/cleaned_text_final`	Cleans random dataset
`src/corpus/clean_text.py --input data/cleaned_text --output data/cleaned_text_final`	Cleans fixed dataset
`src/corpus/embed.py`	Generates dense embeddings and stores them in `.jsonl` format
`src/corpus/bm25_embed.py`	Builds BM25 index (`bm25_index.json`)
`src/corpus/merge_embeddings.py`	Merges fixed and random embeddings into `data/embeddings_merged.jsonl`

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2. Hybrid RAG System

The retrieval system combines **Dense**, **Sparse**, and **Hybrid (RRF)** strategies.

Retrieval & Generation

Script	Description
`src/retrieval/dense.py`	Dense vector retrieval using sentence embeddings
`src/retrieval/sparse.py`	Sparse keyword-based retrieval using BM25
`src/retrieval/hybrid.py`	Hybrid retrieval combining dense and sparse results
`src/retrieval/hybrid.py`	Reciprocal Rank Fusion (RRF) implementation
`src/retrieval/generator.py`	LLM-based response generation
`src/retrieval/test.py`	Tests retrieval and outputs RRF scores for all chunk IDs
`streamlit run app/app.py`	Launches the interactive user interface

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3. Automated Evaluation Framework

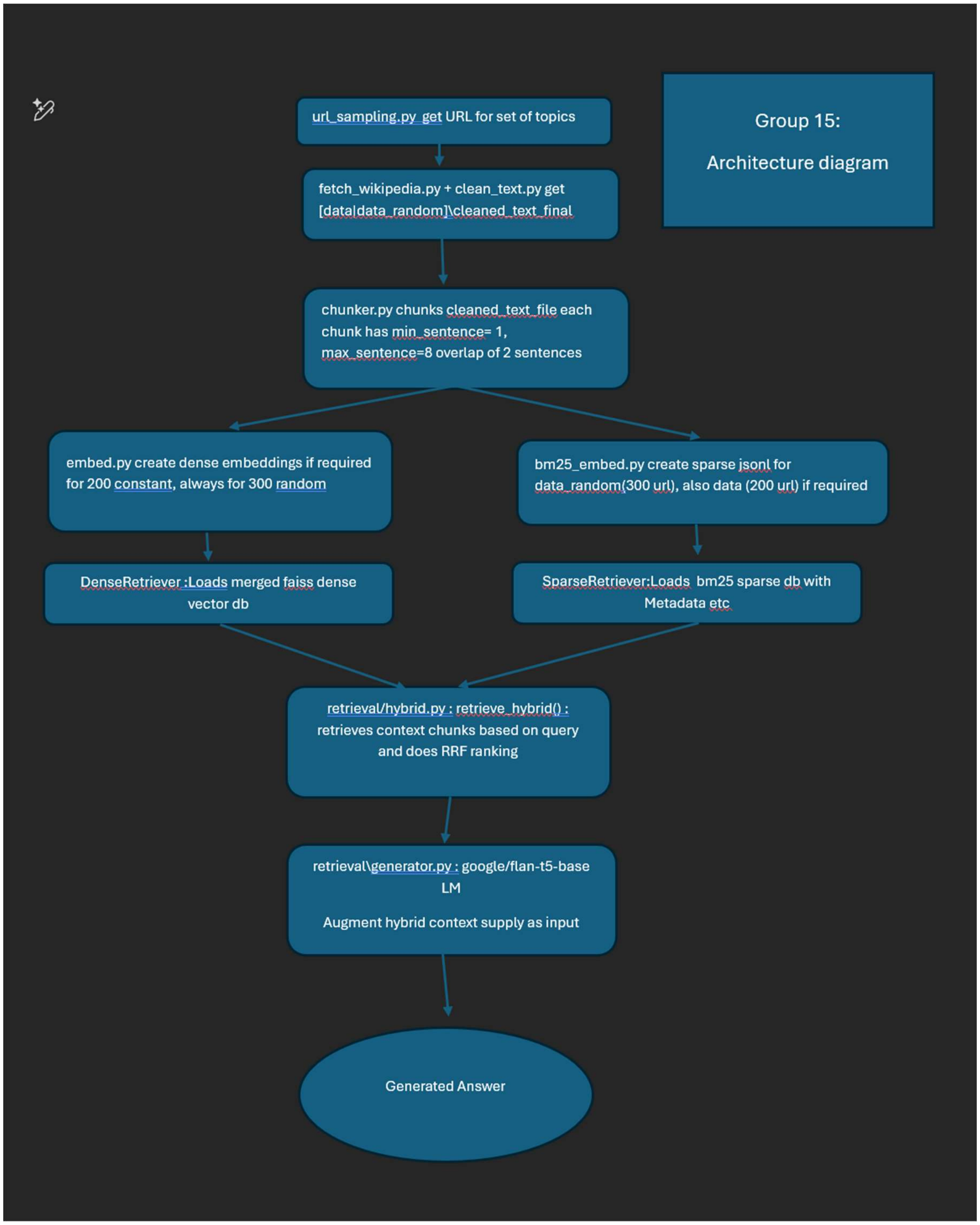
Automated evaluation is performed on **100 generated questions**.

Evaluation Components

Script	Description
`src/retrieval/qgenerator.py`	Automated question generation from Wikipedia corpus
`src/evaluation/eval_MRR.py`	Mean Reciprocal Rank (MRR) evaluation (URL-level)
`src/evaluation/semantic_similarity.py`	Semantic similarity evaluation (answer quality)
`src/evaluation/ablation.py`	Ablation study (Dense vs Sparse vs Hybrid)

End-to-End Execution

All steps—from data collection to evaluation—can be executed using a single script (`run.sh`). Also in readme.md



Local Web Application User Interface Initialization

1. Clone https://github.com/AGDevOps78/rag-hybrid-wiki.git
2. At vscode prompt change directory to path within rag-hybrid-wiki\app
3. Ensure streamlit package is installed, if not run !pip install streamlit

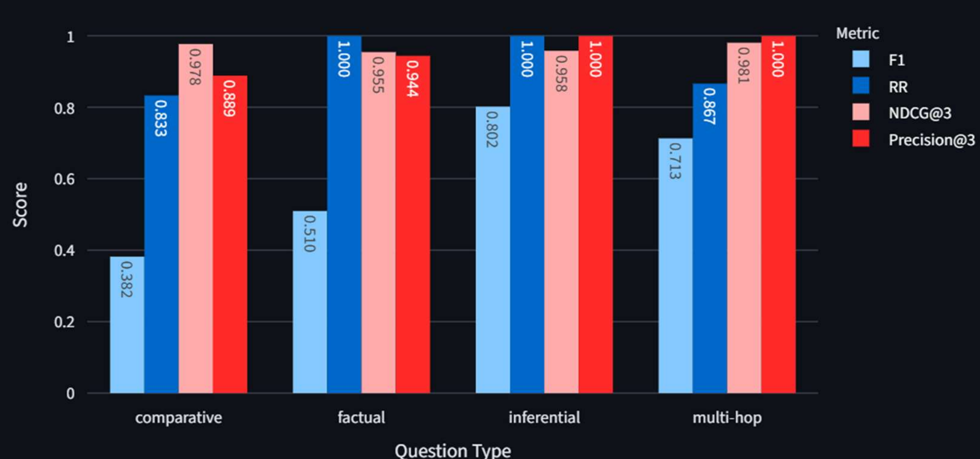
`Run command : streamlit run app/app.py` Launches the interactive user interface

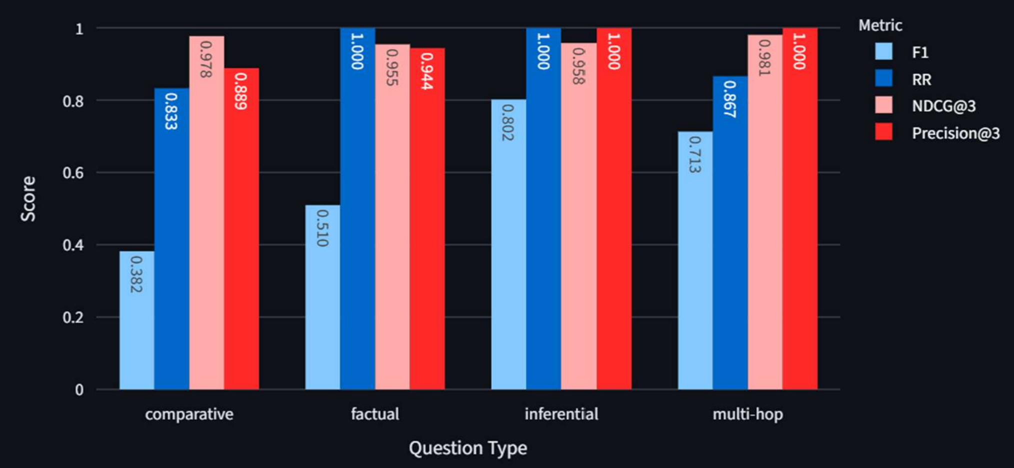
Group 15 Hybrid RAG over Wikipedia

	ID	question	ground_truth
0	1	What are the organelles that eukaryotic cells have?	The cell nucleus
1	2	What is the mechanical advantage of a screw thread?	The mechanical
2	3	What is anti-incumbency?	Anti-incumbenc
3	4	Why is limerence associated with decreased satisfaction?	As romantic obs
4	5	What is the most important role for uranium and thorium fuel precursors synthesis?	Radiochemistry,
5	6	What is the term for the services that are provided by specialist teams of providers in	Tertiary care me
6	7	What is the biological phenomenon of triboluminescence?	Triboluminescer
7	8	What would benefit the well being of older generations?	This would bene
8	9	What is the difference between the two types of literature?	The digital era h
9	10	What does the LinkML schema tries to anchor the meaning of free text strings by esta	Each element in

{'mean_f1': 0.5366009338067843, 'mean_rr': 0.93, 'NDCG@3': 0.9790368175058698, 'mean_semantic_similarity': 0.7067721603065729, 'mean_response_time': 14.256030614376067, 'Precision@3': 0.8966666666666664}

Metrics by Question Type





Enter your question

Who was Albert Einstein ?

Top-K per retriever

5

Top-N after RRF

3

Ask

Mean Reciprocal Rank (URL): 1.0000 over 1 questions

Mean NDCG: 0.8403 over 1 questions at top 3 retrieved chunks

Mean Precision@3: 1.0000 over 1 questions

Mean Response Time: 7.66s over 1 questions

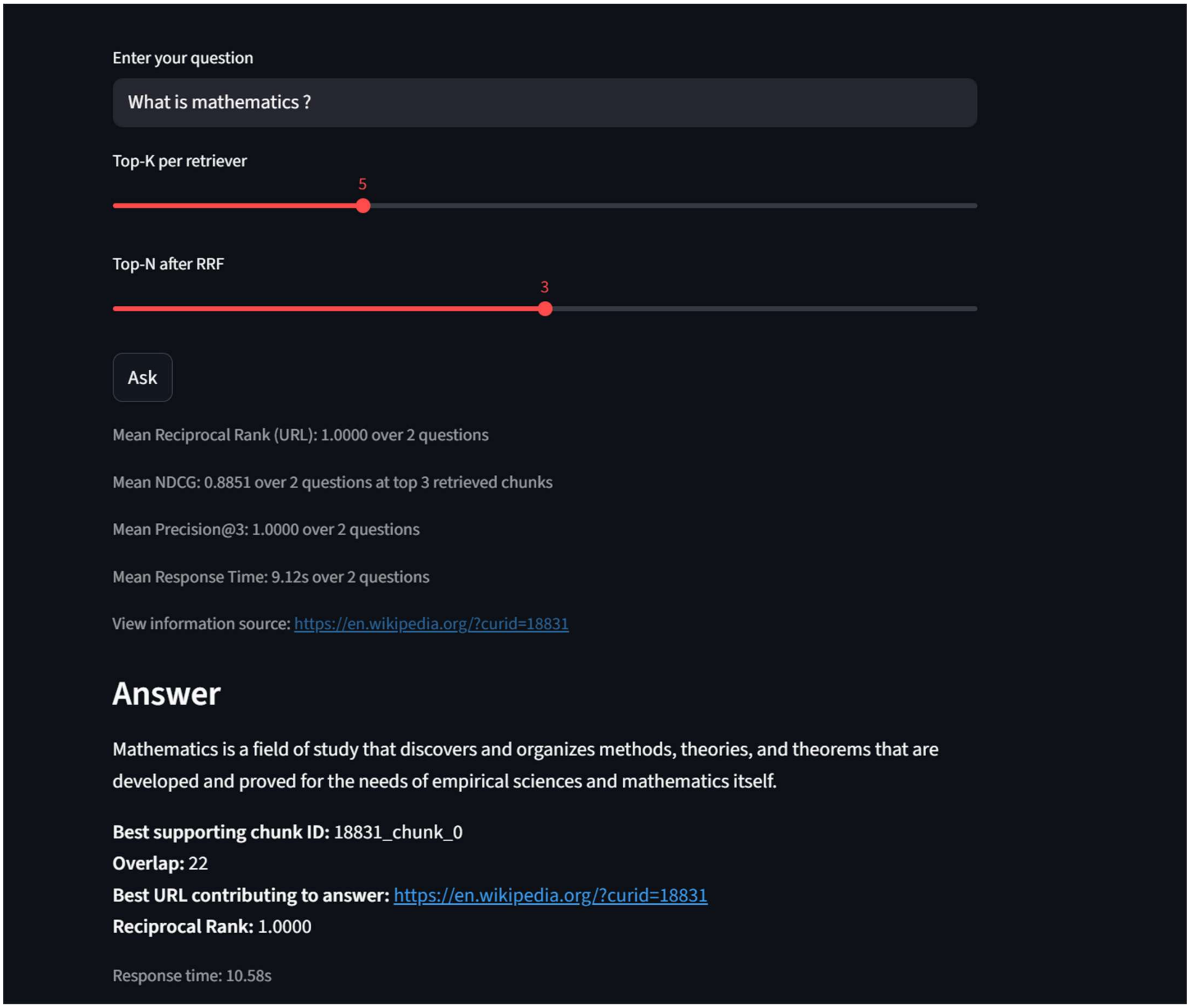
View information source: https://en.wikipedia.org/?curid=65321385

Answer

Albert Einstein was a physicist.

Best supporting chunk ID: 65321385_chunk_11
Overlap: 4
Best URL contributing to answer: https://en.wikipedia.org/?curid=65321385
Reciprocal Rank: 1.0000

Response time: 7.66s



	ID	question	ground_truth	F1	NDCG@3	RR	Semantic Similarity	Overlap	Type	response_time
0	1	What are the organelles that eukaryotic cells have?	The cell nucleus, which contains most of the cell's DNA, or mitochc	1	1	1	1	19	factual	13.1043
1	2	What is the mechanical advantage of a screw thread?	The mechanical advantage of a screw thread depends on its lead, v	1	0.9725	1	1	19	inferential	11.0235
2	3	What is anti-incumbency?	Anti-incumbency is sentiment in favor of voting out incumbent pol	1	0.9778	1	1	24	factual	11.4869
3	4	Why is limerence associated with decreased satisfaction?	As romantic obsession continues inside a relationship over a longe	0.2063	1	1	0.4537	10	comparative	11.3113
4	5	What is the most important role for uranium and thorium fuel precursor	Radiochemistry, radiation chemistry and nuclear chemical enginee	1	0.9152	0.5	1	38	comparative	14.5215
5	6	What is the term for the services that are provided by specialist teams of	Tertiary care medical services are provided by specialist teams of p	0.7143	1	1	0.961	19	factual	11.4127
6	7	What is the biological phenomenon of triboluminescence?	Triboluminescence is thought to be controlled by recombination o	0.5091	1	1	0.789	26	multi-hop	13.7613
7	8	What would benefit the well being of older generations?	This would benefit the well being of older generations, as well as c	0.2143	1	1	0.4116	51	comparative	15.4086
8	9	What is the difference between the two types of literature?	The digital era has blurred the lines between online electronic liter	0.2	0.9725	0.5	0.1411	21	comparative	10.3585
9	10	What does the LinkML schema tries to anchor the meaning of free text st	Each element in the schema can be assigned URIs from existing vo	0.0784	1	1	0.5769	13	factual	10.8961
10	11	What is the most important factor in art valuation?	Information available from internet-based art sale history databas	0.1538	1	1	0.5485	8	factual	9.5744
11	12	Why does the mass media have grown as an institution?	The mass media's current, fragmented, high-choice environment h	1	0.905	0.3333	1	28	multi-hop	11.9417
12	13	What was the name of the person who served as both a student and inst	Stanley Lefevre Krebs	0.3529	1	1	0.7477	13	comparative	9.8093
13	14	What is the risk that the code is broken?	The code is tested so many times (as it is used every time a record	0.9826	1	1	0.9934	74	inferential	22.605
14	15	What is the risk of gestational diabetes associated with classical CAH?	Classical CAH is associated with higher risk of gestational diabetes	1	0.9514	1	1	10	inferential	9.2552
15	16	What is the code used to do?	Every time a record is stored or retrieved, the code has to search fo	0.0759	1	0.5	0.2245	16	factual	13.1611
16	17	What can cause vision with both eyes to be worse than with one eye alo	Binocular interaction occurs when there is an interaction between	0.5	1	1	0.9119	29	multi-hop	13.396
17	18	Why is Turner's work closely linked to Durkheim's seminal work in The R	Turner's work is closely linked to Durkheim's seminal work in The I	0.2424	0.7602	1	0.0528	9	comparative	10.25
18	19	What can be seen as serious threats to the objective understanding of a	Marketing of memory by the culture industry and its instrumentali	0.6552	1	1	0.6671	26	multi-hop	11.3773
19	20	What is the most important role in the manufacture of integrated circuit	In semiconductors it plays an important role e.g. in trapping impur	0.7429	1	1	0.6837	19	multi-hop	10.2436

```
Mean F1: 0.5366, Mean RR: 0.9300, Mean NDCG: 0.9790, Mean Precision@3: 0.8967
   ID                                       question                                       ground_truth        F1    NDCG@3   RR  Semantic Similarity  Precision@3         Type  response_time
0   1  What are the organelles that eukaryotic cells ...  The cell nucleus, which contains most of the c...  1.000000  1.000000  1.0             1.000000          1.0      factual      10.931731
1   2  What is the mechanical advantage of a screw th...  The mechanical advantage of a screw thread dep...  1.000000  0.972504  1.0             1.000000          1.0  inferential       9.978398
2   3                           What is anti-incumbency?  Anti-incumbency is sentiment in favor of votin...  1.000000  0.977781  1.0             1.000000          1.0      factual      10.802758
3   4  Why is limerence associated with decreased sat...  As romantic obsession continues inside a relat...  0.065574  1.000000  1.0             0.529669          1.0  comparative       9.716367
4   5  What is the most important role for uranium an...  Radiochemistry, radiation chemistry and nuclea...  1.000000  0.915151  0.5             1.000000          1.0  comparative      13.180985
```

Evaluation Matrix:

answer-quality evaluation : Set of 100 questions generated with randomly selected chunks and ground truth answers generated by the same SLM (Google-FLAN)

1.**Semantic similarity** between predicted and gold answer using cosine similarity of sentence embeddings from a pre-trained model (all-MiniLM-L6-v2).

This captures the overall semantic closeness of the generated answer to the ground truth answer, beyond just token overlap. It can recognize when two answers are saying the same thing in different words, which is important for evaluating generative models where exact token matches may be less common.

This works by encoding both the predicted answer and the gold answer into dense vector representations using the SentenceTransformer model, and then computing the cosine similarity between these two vectors.

Works better than simple token overlap metrics like F1 because it can capture semantic equivalence even when there are few or no shared tokens, as long as the overall meaning is similar.

However, it may not be as interpretable as token-based metrics and can sometimes give high similarity scores to answers that are semantically related but not actually correct. Therefore, it's best used in conjunction with other metrics like F1 for a more comprehensive evaluation

More useful for larger answers where there is more content to capture semantic meaning, and less useful for very short answers where token overlap may be more indicative of correctness.

:param pred: Answer generated by the RAG system for a given question from K-chunks. This is generated by a LLM (Google Flan-T5-Base) using retrieved K chunks as context.

:param gold: ground truth from a set of questions with known answers. Questions are generatred using SLM (Google Flan-T5-Base) supplying random retrieved chunks as context

ground truth answers are answers generated by the same model (Flan-T5-Base) with the same question but with gold retrieved chunks as context. This way we have a "best possible answer" for each question

that is still generated by the same model and not human annotated, which would be unfairly harsh.

2. F1 score:

F1 score between ground truth and predicted answer based on token overlap after normalization. (answer-level evaluation.

:param pred: Answer generated by the RAG system for a given question. This is generated by a LLM (Google Flan-T5-Base) using retrieved K chunks as context.

:param gold: ground truth from a set of questions with known answers. Questions are generated using SLM (Google Flan-T5-Base) supplying random retrieved chunks as context

ground truth answers are answers generated by the same model (Flan-T5-Base) with the same question but with gold retrieved chunks as context. This way we have a "best possible answer" for each question

that is still generated by the same model and not human annotated, which would be unfairly harsh.

The random questions and answers generated ensure no human bias, and when compared with the RAG generated answer, we can see how close the RAG system gets to the best possible answer at k chunks that the model can generate given perfect retrieval.

Overlap of tokens is computed after normalization which includes: Lowercasing, stop word removal, stemming, and punctuation removal. This is to ensure that we are comparing the core content of the answers rather than surface-level differences.

Retrieval level: Given the predicted answer and chunks check how relevant and grounded the answers are with the retrieved chunks . Higher score means more grounded, less chance of hallucination.

Answer quality: **Average of Semantic similarity and F1 score for answer quality gives a good understanding of answer quality, with respect to ground truth, also avoids too much penalization from F1 score for not using exact word**

1. NDCG@K based on relevance scores computed from token overlap between the generated answer and the retrieved chunk text. (retrieval-level evaluation)

Advantage over precision@k is that it gives more weight to chunks that are more relevant (have higher token overlap) rather than treating all supporting chunks equally. That is NDCG@K score also gives feedback of the ranking quality.

This way we can capture not just whether a chunk supports the answer, but how strongly it supports it based on content overlap.

Sort by rank (in case input isn't sorted)

1. remove stopwords from answer and chunk text

2. use PorterStemmer to stem tokens

2. compute relevance score based on overlap

3. compute DCG and IDCG

2**. precision@k** for supporting chunks in the top-K retrieved chunks.(retrieval-level evaluation)

Here we check how many of the top-K retrieved chunks are actually supporting the answer (based on token overlap) and compute precision accordingly.(count of supporting chunks in top-K / K).

This answers the question “Are all the chunks returned relevant?” if yes its 1. It does not consider the ranking of the chunk and treats all the chunks equally. We use this to check especially at lower K (k<5), if this is less, then it’s possible that 1 or more returned chunks are totally irrelevant

:param ranked_chunk_ids: List of chunk IDs in descending order of relevance.

:param supporting_chunk_ids: Set of chunk IDs that support the answer.

:param k: Number of top chunks to consider for precision calculation.

Why choose 2 each for answer quality and retrieval/grounding?

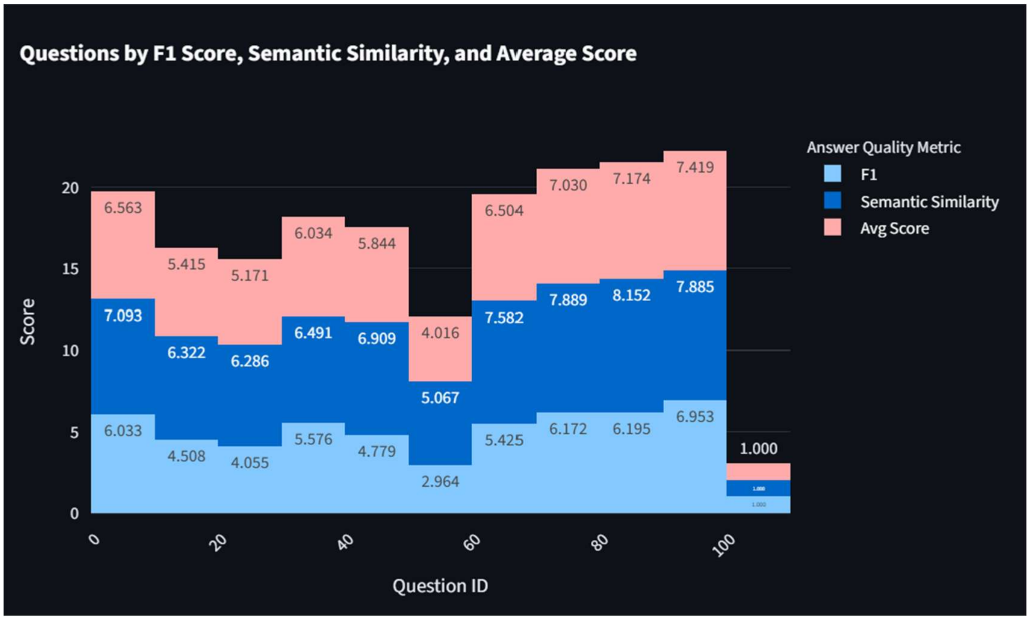
For answer quality, while F1 using token semantic “word for word match” is fast, and accurate for smaller sentences, If the 2 answers being compared, have large differences in word count, even though similar it returns lower score.

Word embedding based Semantic similarity ensures, similar meaning words are not penalized.

Averaging the 2 seems to give a better estimate of answer quality.

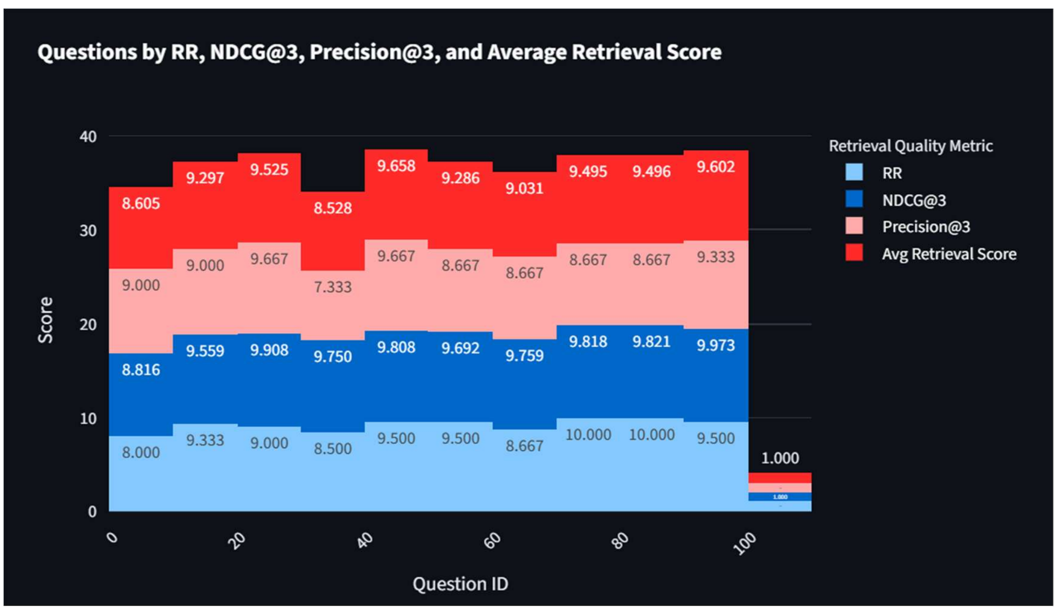
For Retrieval quality , we use precision@k to understand the overall quality of retrieved k-chunks. The more chunks we retrieve and supply to SLM involves more cost in terms of time, resource and tokens used. Given different values of K we try to determine at what K precision tends to 1. Here it seems 3 -5 is a good number

While precision @k gives information about overall retrieval relevance, it does not account for the ranking, which NDCG@k does, this helped, when tweaking the RRF calculation especially for hybrid retrieval



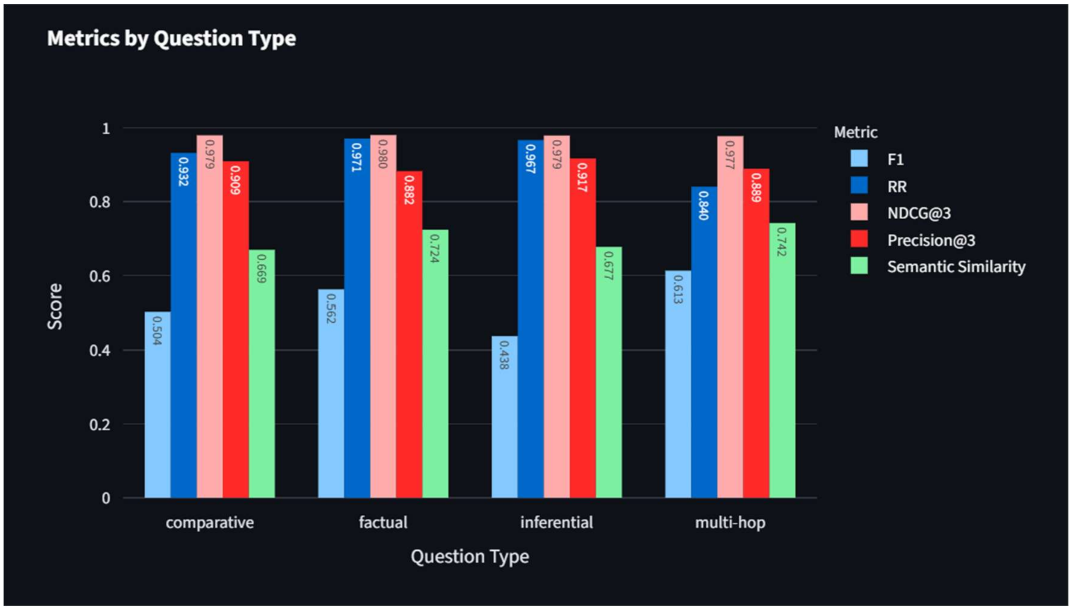
Interpretation of Answer Quality Metric (Avg of F1 and Semantic Similarity w.r.t Golden Answer)

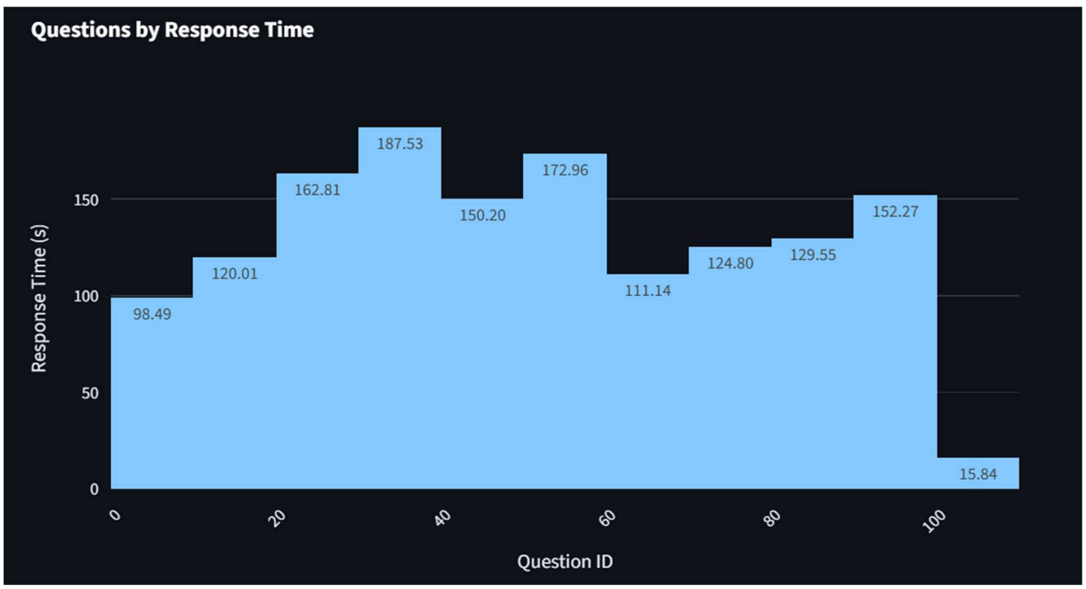
Histogram view by ID, for group of 10 questions, question set between 50-60 has poor answer quality. Highest being at the end, 90-100 seem to return high answer quality average of 0.7. Here the average gives equal weightage to both F1, which is actual word match, Semantic similarity, tries to get similar meaning, where not exactly the same word exists. F1 score is generally lower. So we consider Answer quality avg score > 4/10 as good, >7/10 as excellent



Interpretation of retrieval quality results

Histogram view of retrieval quality results. MRR by URL for 1st 10 questions at 0.8 is the lowest. Precision is lowest at 0.733 for question set 30-40. NDCG is at 0.975 for the same set, so it suggests that relevance is higher in the top 1 or 2 ranked chunks, and some lower ranks are not relevant, ***So we default to 3 chunks*** for quick and relatively accurate retrieval





Mean Response Time of 14 seconds, highest being 18.7 seconds for question set 30-40

Ablation:

We start with initial k = 10 for hybrid, dense and sparse, then try reducing k to get optimised retrieval with low latency. Arrive at k hybrid = 3, k dense and k sparse =3. Essentially top 3 of dense and sparse are RRF ranked, and top 3 RRF ranked chunks are used to generate answers in the final hybrid solution

Ablation result at k =10

```
=== Ablation Results @ k = 10  ===
Dense  : {'MRR': 0.9333333333333332, 'NDCG@10': 0.95, 'SemanticSim': 0.39311151653528215, 'Precision@10': 0.72, 'AnswerF1': 0.05821686519479389}
Sparse : {'MRR': 0.95, 'NDCG@10': 0.9630929753571458, 'SemanticSim': 0.47859506998211143, 'Precision@10': 0.67, 'AnswerF1': 0.13520022068409165}
Hybrid : {'MRR': 0.95, 'NDCG@10': 0.9109061017692044, 'SemanticSim': 0.4347253903746605, 'Precision@10': 0.8, 'AnswerF1': 0.11282309843047669}
```

Ablation result at k = 5

```
=== Ablation Results @ k = 5  ===
Dense  : {'MRR': 0.9333333333333332, 'NDCG@5': 0.95, 'SemanticSim': 0.464617820084095, 'Precision@5': 0.7999999999999999, 'AnswerF1': 0.14769054940532017}
Sparse : {'MRR': 0.95, 'NDCG@5': 0.9630929753571458, 'SemanticSim': 0.5245239654555917, 'Precision@5': 0.8600000000000001, 'AnswerF1': 0.19631133179520277}
Hybrid : {'MRR': 0.95, 'NDCG@5': 0.930140885812723, 'SemanticSim': 0.4794798880815506, 'Precision@5': 0.7999999999999999, 'AnswerF1': 0.13174564505123304}
```

Ablation result at k=3

For k hybrid =3 and k dense, k sparse at 3 as well, while MRR and NDCG values are similar

```
=== Ablation Results @ k = 3  ===
Dense  : {'MRR': 0.9333333333333332, 'NDCG@3': 0.95, 'SemanticSim': 0.7194063812494278, 'Precision@3': 0.8666666666666668, 'AnswerF1': 0.5194903112760255}
Sparse : {'MRR': 0.95, 'NDCG@3': 0.9630929753571458, 'SemanticSim': 0.7104761719703674, 'Precision@3': 1.0, 'AnswerF1': 0.4742231294862873}
Hybrid : {'MRR': 0.9, 'NDCG@3': 0.986543638981406, 'SemanticSim': 0.734885087609291, 'Precision@3': 0.9666666666666666, 'AnswerF1': 0.5719256854256854}
```

If we keep k hybrid = 3, k sparse and k dense at 3 , the semantic similarity and AnswerF1 does slightly better. So we use these values for our hybrid retrieval solution. Also precion@3 and NDCG@3 is the highest

Adversarial Testing (input validation):

Basic check for valid question/query input implemented.

Implementation at (src\evaluation\eval_MRR.py) `is_valid_question_strict`

1. Sematic check for keywords, question marks etc to detect basic issues

2. qchecker based on google/flan-t5-base" with following prompt: (src\retrieval\qchecker.py)

If the question is invalid it fails in 1. Itself, else we pass it to LM. FLAN is being used with STRICT command here, however if for some VALID question we see it return “no” added it to VALID example in the prompt, a few example as below

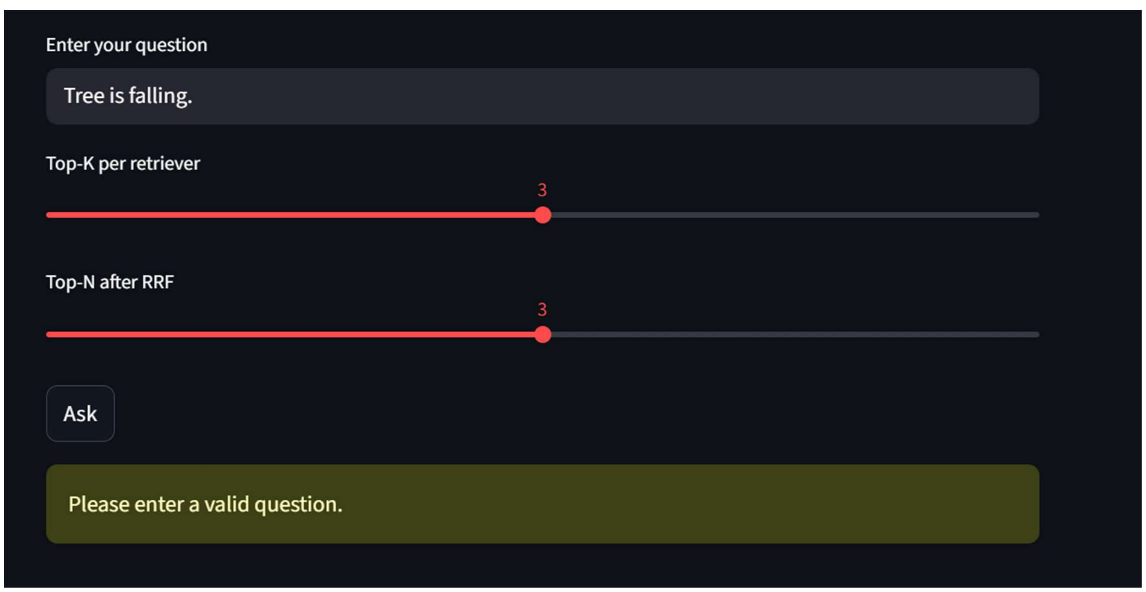
```
 "You are a STRICT classifier. Your job is to determine whether the text is a well-formed question.\n\n"
        "A well-formed question MUST:\n"
        "- Ask for information\n"
        "- Be interrogative in meaning\n"
        "- Not be a greeting, statement, or fragment\n\n"
        "- Start with a question word (e.g. what, why, how) or an auxiliary verb (e.g. is, can)\n"
        "- Starts with Explain, Define, Compare, or Contrast are also valid question starters.\n\n"
        "Examples of valid questions:\n"
        "- What is hybrid RAG?\n"
        "- How does BM25 scoring work?\n\n"
        "- Explain the concept of dense retrieval.\n" \
        "- Define the term 'tokenization' in NLP.\n" \
        "- Compare and contrast dense and sparse retrieval methods.\n\n"
        "Examples of invalid questions:\n"
        "- Hello world ?\n"
        "- The quick brown fox\n"
        "- How are you doing today?\n"
        "- Greetings, how are you?\n"
        "- Hello, can you help me?\n\n"
        "STRICTLY Respond with ONLY YES or NO.\n\n"
        f"Text: {query}"
```

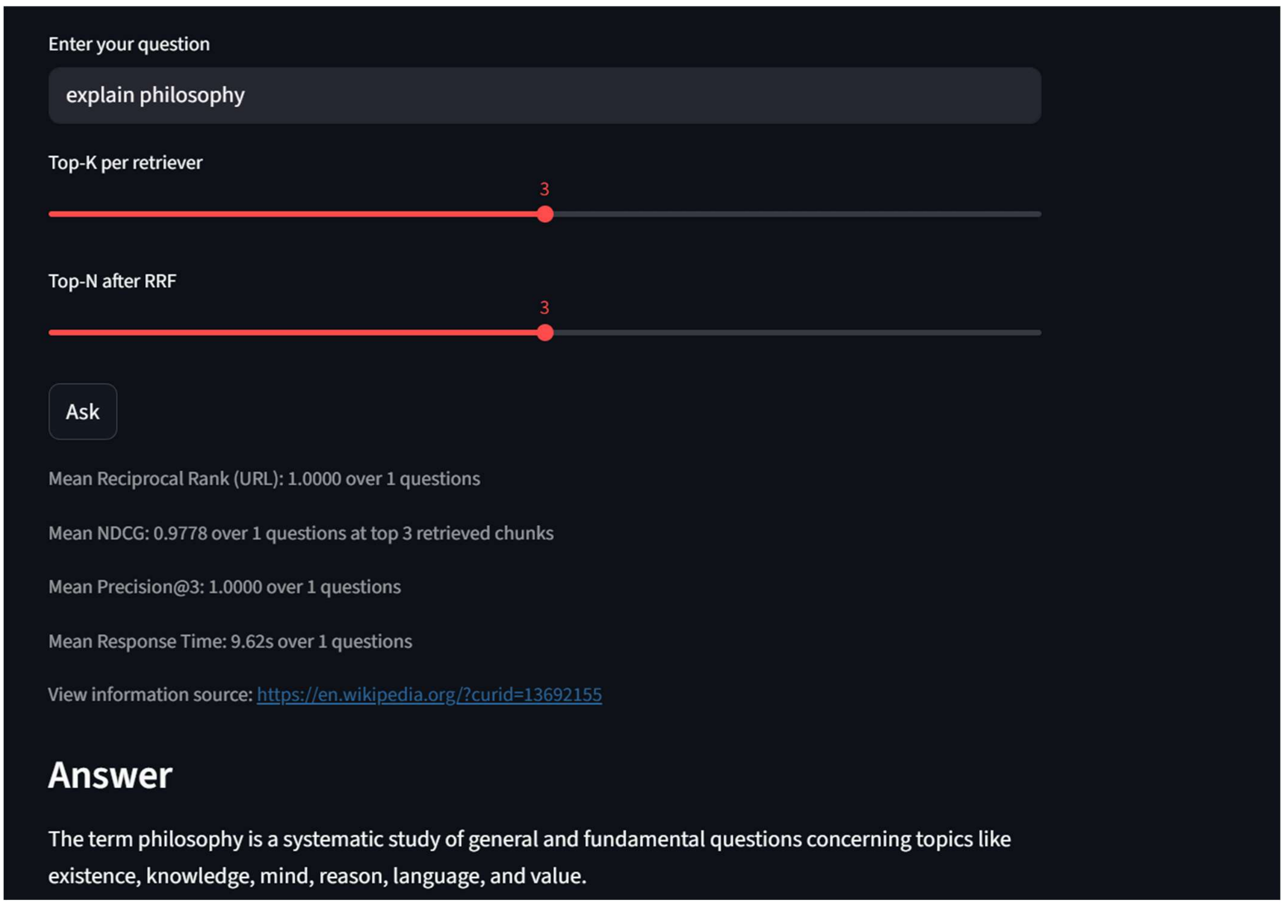
```
Basic checks passed for question: 'How are you doing ?'
LLM Question Check Response: 'no' for query: 'How are you doing ?'
```

Incase for some valid questions the LM responds no, then add to the valid prompt. Also from testing if some invalid questions are not detected, adding them in the INVALID example prompt helps the LM identify it as invalid

```
Basic checks passed for question: 'What is Mathematics?'
LLM Question Check Response: 'valid' for query: 'What is Mathematics?'
```

```
Basic checks passed for question: 'What does the LinkML schema tries to anchor the meaning of free text strings by establishing identit
y via resolvable URIs?'
LLM Question Check Response: 'yes' for query: 'What does the LinkML schema tries to anchor the meaning of free text strings by establis
hing identity via resolvable URIs?'
```





LLM as Judge implementation:

Due to resource constraint we attempted to implement our own LLM assisted automatic judging model as follows

src\evaluation\llm_as_judge.py :

llm_as_judge using google/flan-t5-base , Generate n questions based on random chunks, with sample answer.

Using the n evaluation questions, we supply these to our hybrid retrieval system retrieve_hybrid(query, dense, sparse, top_n=3), and use the retried chunks to augment the supplied question and supply to google/flan-t5-base to generate responses.

These generated responses are evaluated for semantic similarity, by calculating cosine similarity between word embeddings between golden answer and generated response.

Also NDCG value for the retrieved chunks to generate response is calculated.

Score for response is semantic_score*NDCG_score.

For n responses the average "score" is calculated.

```
LLM as a judge generating 5 evaluation questions
Generating 5 Q&A pairs from 220 chunks...
Q output:
What is the relationship between diplomatic and military history?
```

```
Semantic similarity between generated answer and ground truth: 0.6449
NDCG@3 score for retrieved chunks against ground truth: 1.0000
good response score = 0.6449072360992432
avg score for 5 question(s) = 0.7266289883695236)
scores: [1.0, 0.8600201606750488, 0.18215681612491608, 0.94606072894841, 0.6449072360992432]
LLM as a Judge verdict:
good response score = 0.7266289883695236 for 5 questions evaluated
```

```
(pfds) PS C:\Users\anshg\gitrepo\rag-hybrid-wiki> python -m src.evaluation.llm_as_judge
Setting EMB_PATH to: C:\Users\anshg\gitrepo\rag-hybrid-wiki\data\embeddings_merged.jsonl
Loaded dense and sparse retrievers.
Loaded 220 random chunks for question generation.
LLM as a judge generating 10 evaluation questions
Generating 10 Q&A pairs from 220 chunks...
```

```
avg score for 10 question(s) = 0.7899322338525517)
scores: [0.9104296565055847, 0.859740674495697, 0.7234259843826294, 0.9723924398422241, 0.8575224876403809, -0.0, 0.6256372332572937,
LLM as a Judge verdict:
good response score = 0.7899322338525517 for 10 questions evaluated
```